Austin Meadows

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SUMMARY

PhD-trained computational scientist with a decade of experience turning complex oncology questions into reproducible statistical code, analytic datasets, and real-world evidence. Cancer-research background spans hepatocellular carcinoma, melanoma, and immuno-oncology, including a Nature Communications consortium benchmark of statistical cell-type deconvolution methods. Fluent in R (tidyverse) and Python with deep multi-omics, statistical analysis, and machine-learning experience across 8 peer-reviewed publications and **$4.4M** in funded research; translates analysis specifications into descriptive and inferential statistics, visualizations, and version-controlled pipelines for cross-functional clinical and translational teams.

SKILLS

Scientific programming: R (tidyverse), Python (pandas, NumPy), SQL, Bash, MATLAB, C/C++, SPSS, Prism

Machine learning: statistical modeling, PyTorch, transformers/LLMs, CNNs, GNNs; model training, fine-tuning, evaluation, backpropagation/autograd

Biomedical & Real-World Data: Multi-omics integration (RNA-seq, ATAC-seq, CUT&RUN/Tag, ChIP-seq), clinical-translational diagnostics, genomics, epigenetics, immunology, oncology, cell-type deconvolution

Infrastructure: Git/GitHub, HPC (IBM LSF), Docker, Conda; scalable & reproducible pipelines;

Wet Lab: NGS library prep & QC, flow cytometry, immunofluorescence/microscopy, qPCR, ELISA, western blot, cell culture, mouse work

Leadership: Cross-functional & cross-institutional research, grant writing, manuscript & technical writing, scientific presentation, lab management

EXPERIENCE

Founding Scientist — Immunome AI Biotechnologies Sep 2025 – Present
New York, NY

- Co-developed iMEMORY, an AI platform generating clinically relevant immune-profiling and drug-target-discovery insights from heterogeneous patient data, advising on platform design and biological interpretation of model outputs.
- Nominated in-silico therapeutic gene targets and supplied doctoral datasets used in a cross-species validation of the platform's predictions.
- Produced investor-facing scientific materials and presentations that translated complex model outputs for non-technical stakeholders.

PhD Candidate, Biomedical Sciences & Public Health — Icahn School of Medicine at Mount Sinai / Instituto de Salud Carlos III Dec 2023 – Dec 2026 (est.)
New York, NY / Madrid, Spain

- Designed and coordinated a multi-omics study of trained immunity, generating and analyzing ATAC-seq, RNA-seq, and CUT&RUN data through reproducible, version-controlled R and Python pipelines.
- Built the first ex-vivo model and the first clinical-setting diagnostic for dialysis-mediated trained immunity, bridging bench data to a translational clinical readout.
- Authored a publication-grade manuscript end to end, including the statistical analysis plan, descriptive and inferential statistics, and all figures.

Associate Researcher — Icahn School of Medicine at Mount Sinai Dec 2021 – Sep 2025
New York, NY · Tisch Cancer Institute

- Generated the preliminary data underpinning successful NIAID and NCI grant applications totaling **$4.4M**.
- Served as genomics and computational-biology subject-matter expert for multiple labs across the Tisch Cancer Institute, MGH, and WashU, delivering analyses in a cross-institutional environment.

- Elucidated the mechanism of action of novel hepatocellular-carcinoma chemotherapy WNTinib by designing, conducting, and analyzing the epigenomic experiments (*Rialdi et al., Nature Cancer, 2023*).
- Identified two protein complexes whose disruption arrests melanoma cell-cycle progression, enabling a new epigenetic strategy for drug-resistant melanoma (*Jostes et al., Genes & Development, 2024*).
- Developed the first nanoparticle biologic to attenuate allograft fibrosis by silencing immune training (*Mineura et al., Am. J. Transplantation, 2026*).

Sequencing Core Technician — Girihlet, Inc. Feb 2021 – Dec 2021
New York, NY

- Engineered a Python interface that automated creation and submission of NGS protocols to a liquid-handling robot, eliminating manual workflows and saving tens of thousands of dollars in annual overhead.
- Delivered tailored NGS services and experimental consultation to NYC-area biomedical research clients.

Research Technician — NYIT College of Osteopathic Medicine Nov 2018 – Jul 2020
Long Island, NY

- Built an in-silico cell-type deconvolution method estimating tumor composition from bulk RNA-seq; benchmarked against community methods in a Nature Communications consortium study (*White et al., 2024*).
- Extended the CPA algorithm with a machine-learning post-processing layer (CNN + autoencoder) for single-molecule localization microscopy, trained on a synthetic dataset generated from a physical model of the microscope.

EDUCATION

Ph.D., Biomedical Sciences & Public Health Dec 2026 (est.)
Icahn School of Medicine at Mount Sinai / Instituto de Salud Carlos III

B.E., Biomedical Engineering · B.S., Applied Mathematics & Statistics May 2018
Stony Brook University

SELECTED PUBLICATIONS

Mineura, K., Liu, C. R., …, **Meadows, A.**, …, & Schotsaert, M. (2026). Myeloid-avid mTOR-inhibiting nanobiologic attenuates allograft fibrosis after lung transplantation. American Journal of Transplantation. https://doi.org/10.1016/j.ajt.2026.05.2303

Meadows, A., Dimitrov, A., … & Ochando, J. (2026). Investigating the induction of trained immunity in myeloid-derived monocytes following extracorporeal circulation. *Manuscript in progress*.

White, B. S., de Reyniès, A., Newman, A. M., **Meadows, A.**, et al. (2024). Community assessment of methods to deconvolve cellular composition from bulk gene expression. *Nature Communications*, 15, 7362.

Jostes, S., Vardabasso, C., … **Meadows, A.**, … & Bernstein, E. (2024). H2A.Z chaperones converge on E2F target genes for melanoma cell proliferation. *Genes & Development*, 38(7–8), 336–353.

Rialdi, A., Duffy, M., Scopton, A. P., **Meadows, A.**, et al. (2023). WNTinib is a multi-kinase inhibitor with specificity against β-catenin mutant hepatocellular carcinoma. *Nature Cancer*, 4, 1157–1175.

Full publication list (8 peer-reviewed) available at my website, or upon request.

SELECTED PROJECTS

- **microgad** — Built a neural network and reverse-mode autograd engine from scratch in pure Python (no ML frameworks), implementing backpropagation, ReLU/tanh activations, hinge loss, and SGD.
- **nanoGPT** — Built a decoder-only transformer language model from scratch in PyTorch, implementing multi-head self-attention, positional embeddings, and the full training loop.